Project Proposal: Local RAG-Powered Sepsis Clinical Treatment Recommender

Subject: Proposal Idea: Local RAG-Powered Sepsis Clinical Treatment Recommender

1. Introduction & Context

Sepsis remains one of the leading causes of mortality in emergency and critical care settings, where each hour of delayed resuscitation and antimicrobial therapy significantly worsens clinical outcomes. To address this clinical urgency, this proposal introduces a standalone, client-side **Retrieval-Augmented Generation (RAG)** architecture and interactive decision-support system. Purpose-built as a point-of-care sepsis clinical decision support (CDS) assistant for ICU and emergency physicians, the platform streamlines Surviving Sepsis Campaign (SSC) guideline synthesis, reduces cognitive load during time-critical deterioration, and supports rapid, evidence-backed resuscitation and antimicrobial prescribing decisions during clinical consultations.

2. Project Overview & Shared Vision

Our goal is to build a privacy-preserving, web-based clinical decision support tool that maps acute patient physiological and laboratory metrics to established sepsis management guidelines, the **Surviving Sepsis Campaign (SSC) 2021 International Guidelines (1-Hour and 3-Hour Bundles)**, **Sepsis-3 Consensus Definitions**, and a curated **Renal/Hepatic Antimicrobial Dosing Protocol**, using Generative AI .

By combining a browser-first web framework with lightweight local LLMs, a pre-computed guideline vector index, and deterministic clinical rules, our solution achieves three critical objectives:

1. **Absolute Privacy & Local Execution:** Sensitive clinical metrics entered by physicians at the point of care remain entirely on the client's device, running locally in the browser via WebGPU/WebAssembly (WASM) to ensure full HIPAA/GDPR compliance and zero institutional data leakage.
2. **Strict Grounding & Zero Hallucination:** The generative AI relies entirely on local semantic retrieval from verified sepsis literature paired with deterministic physiological thresholds rather than model memory, ensuring every recommendation is directly traceable to medical literature.
3. **Transparent Decision Support:** Physicians receive rapid, verifiable clinical rationales accompanied by explicit citations and confidence markers to support high-stakes clinical judgment.

3. Detailed System Features & App Capabilities

To ensure the application delivers high clinical utility while remaining computationally lightweight in the browser, it incorporates the following core modules:

- **Hybrid Client-Side LLM & Pre-Computed Vector Engine:** Uses a lightweight, quantized open-source model (1B–3B parameters such as `Llama-3.2-1B-Instruct` or `Qwen2.5-1.5B-Instruct` via WebLLM/WebGPU) paired with a pre-computed in-browser vector index (~150–200 curated guideline chunks). Pre-computing embeddings offline reserves 100% of client WebGPU memory for fast bedside LLM synthesis and instantaneous cosine similarity search.
- **Deterministic Organ-Function & Antimicrobial Contraindication Engine:** Designed for physician order review during acute sepsis management, a fast rule-based physiological layer cross-references empiric broad-spectrum antibiotics (e.g., vancomycin, piperacillin-tazobactam, cefepime, meropenem), fluid resuscitation orders, and vasopressors against real-time renal (creatinine/BUN), hepatic (bilirubin/AST), and hematologic (platelets/PTT) thresholds to proactively flag nephrotoxic/hepatotoxic dosing adjustments and fluid overload risks.
- **Interactive "What-If" Resuscitation & Scenario Modeling:** A dynamic treatment-simulation workspace allowing physicians to test hemodynamic and metabolic interventions in real time (e.g., titrating norepinephrine to achieve MAP ≥ 65 mmHg, administering a 30 mL/kg crystalloid bolus, or tracking lactate clearance) to instantly recalculate deterministic SOFA sub-scores and trigger updated SSC bundle guidance.
- **Uncertainty Estimation & Escalation Flagging:** Implements lightweight calibration markers (combining retrieval cosine-similarity thresholds and rule concordance) that explicitly alert the attending physician when atypical presentations, culture-negative shock, or refractory hypotension fall outside standard bundle pathways, prompting infectious disease or critical care escalation.

4. Dataset Integration: PhysioNet Challenge 2019 Dataset

To evaluate and demonstrate our recommender system on realistic, multi-dimensional ICU patient profiles, we are integrating the **PhysioNet / Computing in Cardiology Challenge 2019 Dataset** (*Early Prediction of Sepsis from Clinical Data*).

- **Dataset Characteristics:** Contains multi-center ICU time-series data covering 40,336 patient admissions across 41 total columns: 34 hourly physiological variables (8 vital signs and 26 laboratory values, including MAP, SBP, HR, Temp, RR, O2Sat, FiO2, SaO2, Lactate, WBC, Platelets, Total Bilirubin, and Creatinine), 6 demographic/administrative fields (including `ICULOS` hours in ICU), and the binary Sepsis-3 outcome label (`SepsisLabel`, which transitions to `1` six hours prior to clinical sepsis onset). It is openly accessible without restrictive IRB/CITI credentialing delays.

- **Application in the Project:**
    - *Curated Longitudinal ICU Cohort Extraction:* Extract a benchmark set of 20–25 archetypal hourly patient trajectories across non-sepsis SIRS controls, pre-sepsis deterioration (`SepsisLabel = 0` transitioning to `1`), acute organ dysfunction (computing hourly SOFA sub-scores from `MAP`, `Creatinine`, `Bilirubin_total`, `Platelets`, and `SaO2`/`FiO2`), and septic shock. Include automated unit normalization in the data loader (e.g., converting dataset Lactate from mg/dL to clinical mmol/L alongside raw values).
    - *RAG Pipeline Validation & Stress Testing:* Use this curated 20–25 patient ICU benchmark cohort to rigorously evaluate local retrieval precision across SSC 1-hour/3-hour bundles, renal antimicrobial dosing flags, and inference latency across realistic stages of clinical deterioration.

5. End-to-End System Workflow Example

1. **Physician Input & Bedside Chart Loading:** During an acute ICU evaluation, the physician loads an hourly patient trajectory from the curated PhysioNet 2019 cohort (or enters bedside chart metrics):
    - *Age:* 68 | *Temp:* 38.9 °C | *HR:* 118 bpm | *MAP:* 58 mmHg (SBP 88 mmHg) | *RR:* 26 breaths/min | *Lactate:* 3.8 mmol/L (34.2 mg/dL) | *WBC:* 18.4 × 10^3/μL | *Platelets:* 115 × 10^3/μL | *Creatinine:* 2.1 mg/dL
2. **Local Semantic Retrieval & Organ-Safety Protocols:** The deterministic physiological engine computes the acute SOFA sub-scores and flags hypotension (MAP < 65 mmHg), hyperlactatemia (> 2.0 mmol/L), and acute kidney injury (Creatinine 2.1 mg/dL), while the local vector index retrieves targeted excerpts from the Surviving Sepsis Campaign (SSC) 1-hour bundle and antimicrobial renal dosing tables.
3. **Clinical Synthesis & Prescribing Alerts:** The browser-based LLM synthesizes a physician-facing recommendation with a **High Confidence Score** (direct Sepsis-3 and SSC bundle match), recommending immediate blood cultures, broad-spectrum empiric antibiotics with explicit renal dose adjustment alerts for elevated creatinine (2.1 mg/dL), and a 30 mL/kg IV crystalloid bolus.
4. **Interactive "What-If" Resuscitation Modeling:** The physician uses the scenario workspace to simulate persistent hypotension (MAP < 65 mmHg post-fluid bolus) and evaluates initiating norepinephrine titration alongside repeat lactate monitoring, viewing updated SSC vasopressor pathways and literature citations in real time.

6. 9-Day Project Milestones & Execution Roadmap

To deliver a functional, validated point-of-care MVP within a 9-day timeframe, development is structured into four focused sprints:

- **Days 1–2 (Milestone 1: Guideline Knowledge Base & PhysioNet Cohort Curation):**
    - Parse and chunk high-yield sections of the **Surviving Sepsis Campaign (SSC) 2021 Guidelines** (1-hour/3-hour bundles, fluid resuscitation, vasopressors) and a

5-drug Renal/Hepatic Antimicrobial Dosing Table (vancomycin, piperacillin-tazobactam, cefepime, meropenem, gentamicin) into ~150–200 structured chunks with citation metadata.
    - Pre-compute vector embeddings offline into a lightweight static JSON index for zero-latency browser loading.
    - Extract 20–25 representative hourly ICU patient `.psv` trajectories from the PhysioNet Challenge 2019 dataset with automated unit conversion (e.g., Lactate mg/dL to mmol/L) and hourly SOFA sub-score calculation.
- **Days 3–5 (Milestone 2: Client-Side WebLLM & Hybrid Safety Engine):**
    - Integrate **WebLLM** (targeting `Llama-3.2-1B-Instruct-q4f16_1-MLC` or `Qwen2.5-1.5B-Instruct` via WebGPU) with browser caching (`IndexedDB`) and static cosine-similarity retrieval.
    - Build and unit-test the deterministic JavaScript clinical rule engine to evaluate Sepsis-3 criteria, MAP/Lactate thresholds, and renal/hepatic antimicrobial dose flags in <10 ms before passing structured context to the LLM.
    - Tune structured system prompts and output schemas to enforce strict inline guideline citations and confidence calibration.
- **Days 6–7 (Milestone 3: Physician Point-of-Care UI & "What-If" Workspace):**
    - Build the bedside web dashboard allowing physicians to select a PhysioNet ICU patient profile, scrub through their hourly ICU timeline (`ICULOS`), and inspect live vital/lab trends.
    - Implement interactive "What-If" sliders and intervention toggles (fluid bolus, vasopressor titration, antimicrobial selection) coupled with confidence/escalation badges and inline guideline citations.
- **Days 8–9 (Milestone 4: Clinical Validation, Benchmarking & Documentation):**
    - Benchmark end-to-end retrieval precision, citation faithfulness, contraindication recall, and browser inference latency (tokens/sec and cold/warm start times) across the 20–25 curated PhysioNet sepsis trajectories.
    - Conduct edge-case stress testing (e.g., culture-negative refractory shock and severe acute kidney injury) and finalize system documentation and clinical demonstration workflows
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